

# IHAB TABBARA

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## EDUCATION

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| <b>Washington University</b><br>PhD. Computer Science and Engineering<br>Focus: Machine Learning, Formal Methods, Safe Control                                              | 2024 - 2029 |
| <b>American University of Beirut</b><br>BE Computer and Communication Engineering <b>GPA: 4.0/4.0</b><br>Focus: Machine Learning, Estimation and Detection Theory, Robotics | 2020 - 2024 |
| <b>University of Toronto (Exchange Program)</b><br>Engineering-Science <b>GPA: 85%</b><br>Relevant Courses: Machine Learning - Advanced Algorithms - Control Systems        | 2022 - 2023 |

## WORK EXPERIENCE

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| <b>Murex</b><br><i>Software Engineer</i>                                                                                                                                                                                                                                                                                                                                                                                                          | June 2023 - May 2024           |
| <ul style="list-style-type: none"><li>Designing and developing financial data structures in C++ for pricing and risk analysis</li><li>Managing sprint planning, daily standups, and sprint retrospectives as scrum master</li></ul>                                                                                                                                                                                                               |                                |
| <b>Electrical and Computer Engineering Department - AUB</b><br><i>Lab Assistant - Teaching Assistant</i>                                                                                                                                                                                                                                                                                                                                          | December 2022 - September 2023 |
| <ul style="list-style-type: none"><li>Provided assistance and guidance to over 100 students in electronics and microprocessors within the ECE 410 Lab (System Integration Laboratory)</li><li>Designed new experiments involving controllers using Raspberry Pi, Arduino and MATLAB for the revamped ECE460L (Control Systems Laboratory)</li><li>Created detailed lectures powerpoints and tutorials and assisted in teaching the lab.</li></ul> |                                |

## RESEARCH EXPERIENCE

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| <b>Washington University</b><br><i>Researcher with Prof. Hussein Sibai</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | August 2024 - Present    |
| <ul style="list-style-type: none"><li>Applying machine learning techniques to learn robust control barrier functions for dynamic, real-time systems in vision-based autonomous control</li><li>Designing and learning safe, formally-verified policies and optimization frameworks for autonomous vehicles with unknown control dynamics</li></ul>                                                                                                                                                                          |                          |
| <b>University of Virginia</b><br><i>Research Assistant (Eye Tracking) with Prof. Sara Riggs</i>                                                                                                                                                                                                                                                                                                                                                                                                                             | May 2023 - February 2024 |
| <ul style="list-style-type: none"><li>Implementing an adaptive velocity threshold eye tracking algorithms and bench marking it against other algorithms such as HMM, and Local Nearest Neighbor algorithms</li><li>Worked on developing a software to control functionalities of small vibrating devices (tactors) using tools such as docker, C, Rust, linux</li></ul>                                                                                                                                                     |                          |
| <b>American University of Beirut</b><br><i>Research with Prof. Hussein Hussein (Robotics)</i>                                                                                                                                                                                                                                                                                                                                                                                                                               | September 2022 - Present |
| <ul style="list-style-type: none"><li>Designed and researched modular robots (Kubits) utilizing Electro-Programmable Magnets (EPMs)</li><li>Conducted comprehensive optimization of module parameters, including magnet dimensions, cube geometry, H-bridge configurations, and gear sizing to improve size and power efficiency.</li><li>Investigated electromagnetic force modeling between repelling and attracting robotic modules, contributing to the understanding of modular robotic system interactions.</li></ul> |                          |

## PUBLICATIONS

- Tabbara I, Sibai H. (2025). Learning Neural Control Barrier Functions from Offline Data with Conservatism. arXiv preprint, arXiv:2505.00908v1. **Accepted to ICRA 2026**
- Tabbara, et al. (2025). Designing Latent Safety Filters using Pre-Trained Vision Models. arXiv:2509.14758v1. **Accepted to ICRA 2026**
- Huriot S, Tabbara I, Sibai S. Safe Control using Learned Safety Filters and Adaptive Conformal Inference. **Accepted to googL4DC 2026.**
- Atweh, Jad A., Ihab Tabbara, et al. 2024. Reducing Barriers in Analyzing Eye Tracking Data: The Development of a GUI to Preprocess Eye Tracking Data. **Accepted to IEEE SIEDS 2024**
- Tabbara, et al. (2025). Computing Sound and Accurate Upper and Lower Bounds on Hamilton-Jacobi Reachability Value Functions. arXiv: 2511.15238
- Tabbara, et al. (2025). Statistically Assuring Safety of Control Systems using Ensembles of Safety Filters and Conformal Prediction, arXiv:2511.07899v1.